Improved Ingredients-based Recipe Recommendation Software using Machine Learning-2023 IEEE

- ResNet-V2 architecture
- 33 different food ingredients
- accuracy rate of 89%.
- ResNet50
- input size of 224 x 224 pixels.
- Data Augmentation
- Adam optimizer with a learning rate of 0.001 and learning rate decay.

- **Dataset:** e Food.com dataset from Kaggle [17], that contains thousands of recipes, which offers various details on each recipe including cooking hours,
  o classify the ingredients provided by the user, we got customized food ingredients dataset from GitHub that contains a collection of food ingredient images. The dataset contains 3,500 images of 33 different various food ingredients categories, such as fruits, vegetables, meats, fish, and chicken. We used this dataset as input data for our classification model to recommend recipes based on the user's input.
  The preference dataset used in the system comprises the preferences selected by users during the application's signup process that results in the variety of different recipes for each user. The dataset includes 10 distinct types of preferences, namely: spicy, lactose-free, low protein, high protein, healthy, vegan, very low carbs, high carbs, low cholesterol, and high/low blood pressure.

- 50 epochs using transfer learning and applying early stopping, our proposed CNN-based ResNet50 model achieved a 95.69% accuracy for the training dataset. And 89.94% accuracy on the validation dataset.

Food Recipe Recommendation Based on Ingredients Detection Using Deep Learning:ICCA 2022

ResNet50
32 ingredients: Beef Meat, Bell Pepper, Burger Bun, Burger Patty, Chick pea, Cheese, Chicken, Chocolate, Chocolate Syrup, Donut Bun, Egg, Figure 5: 32 kinds of food ingredients in our dataset. GroundBeef, Hotdog Bun, Ice Cream, Lasagna Noodles, Letus, May onnaise, Milk, Mushrooms, Nachos Chips, Noodles, Onion, Pasta, Pizza Dough, Potato, Rice, Roll Bread, Sandwich Bread, Sausage, Shawarma Bread, Subway Bun, and Tomato.
accuracy of 94%
19 kinds of cooking recipes associated with 32 ingredients

food101, fruit 360, and UECFOOD256 and the data we created. From fruit360, we took the potato and tomato datasets, and from food101, we took the ice cream and onion datasets, and finally, from UECFOOD256, we took the rice, chicken, roll bread, and sausage datasets. In total, we made the dataset which contains a total of 9856 images of 32 different ingredients

resized the images to 224 × 224 pixels and excluded unnecessary objects from the images

RecipeIS—Recipe Recommendation System Based on Recognition of Food Ingredients

36classesof food ingredients; • 36 classes of food ingredients; • 3115imagesintotal; • 3115 images in total; • Fruitandvegetable images; • Fruit and vegetable images; • ImagesinPNGimageformat; • Images in PNG image format; • Different sizes and resolutions; • Different sizes and resolutions; • Imageswithdifferent angles, types of luminosity, and quality. • Images with different angles, types of luminosity, and quality.

92% accuracy on the training dataset (accuracy) and 83%onthe test dataset (val_accuracy).